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ROYAL MELBOURNE INSTITUTE OF TECHNOLOGY
SGS CAMPUS

ISYS2581 – Blockchain Applications and Smart Contracts

Assignment 2

BLOCKCHAIN APPLICATION PROPOSAL

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I. INTRODUCTION

The paradigm is shifting to more sustainable food models as some CSA programs, including GreenHarvest, connect urban consumers with local farmers who offer them regular shares of fresh produce (Visimex 2023). These programs enhance food security, ensure sustainable agriculture practices, and stimulate local economies.

However, the current CSA programs are facing challenges related to manual processes, high administrative costs, and limited visibility into product provenance. The PESTLE-light analysis is conducted (Table 1) to highlight the relevant factors.

Factor	Description
Economic	Farmers face cash flow problems as payments are delayed due to manual reconciliation processes (20-40 days) => Hindering the reinvestment capability and fund accessibility of farmers (Kamath 2018)
Social	Lack of consumer trust in the certification processes, leading to skepticism about product provenance (Kamilaris et al. 2019) => Ongoing skepticism drives consumers to seek alternatives, damaging GreenHarvest's reputation
Technology	Limited automation technologies are integrated into operations, thus heavily relying on manual reconciliation => Hindering the scalability of the program, as operational costs increase with program growth, making the model unsustainable.

Table 1: PESTLE-light analysis of GreenHarvest's CSA

Through PESTLE analysis, these three inefficiencies are not simply daily operational issues; if remain unresolved, they could lead to GreenHarvest's inability to scale, causing loss on a broader scale, affecting both farmers and consumers.

The five whys analysis is utilized to understand in-depth the inefficiencies, identify the root causes, and further recommend the proper solutions

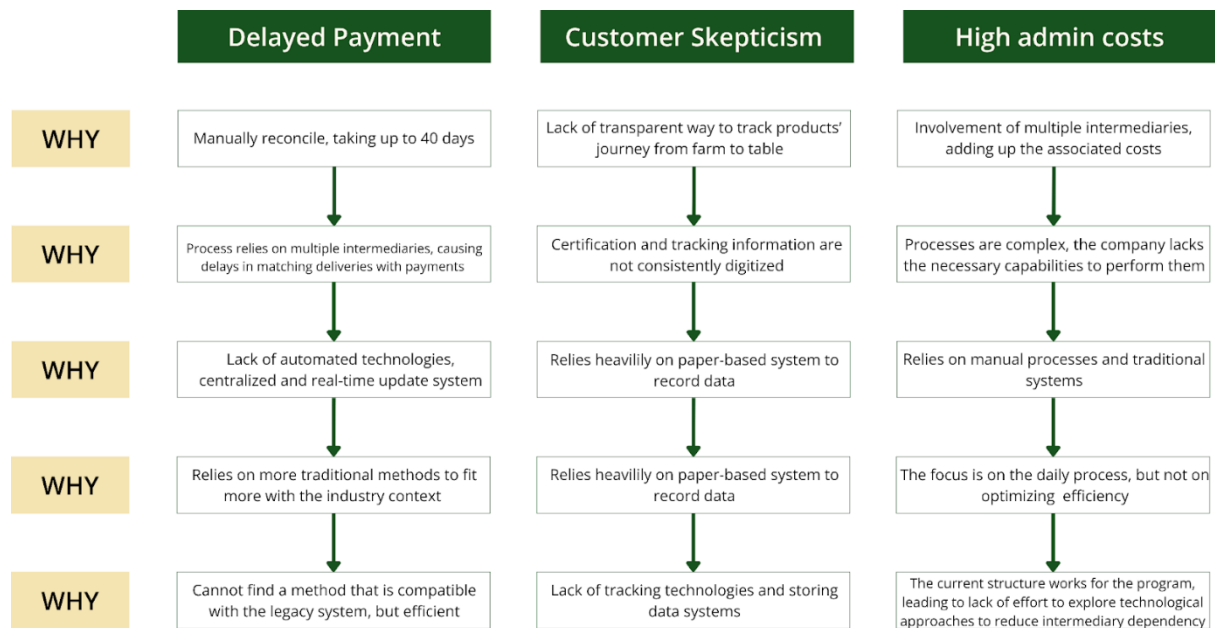


Figure 1: Five whys analysis for three problems that GreenHarvest's CSA program is encountering (Source: Author)

GreenHarvest's inefficiencies stem from the lack of tracking and storing data technologies, leaving operations with redundant manual processes (Figure 2). Therefore, to address this gap, it is necessary to adopt technological solutions that enable automatic fund triggering, certification & product data storage, and reduce reliance on intermediaries. While some CSA programs adopt ERP systems, these still fail to thoroughly resolve payment delays and certification skepticism, highlighting the need for blockchain-based solutions.

II. BLOCKCHAIN INTEGRATION PROPOSAL

The Ten-Step Decision Path helps businesses assess the suitability of blockchain technology for their operations (Pedersen et al. 2019).

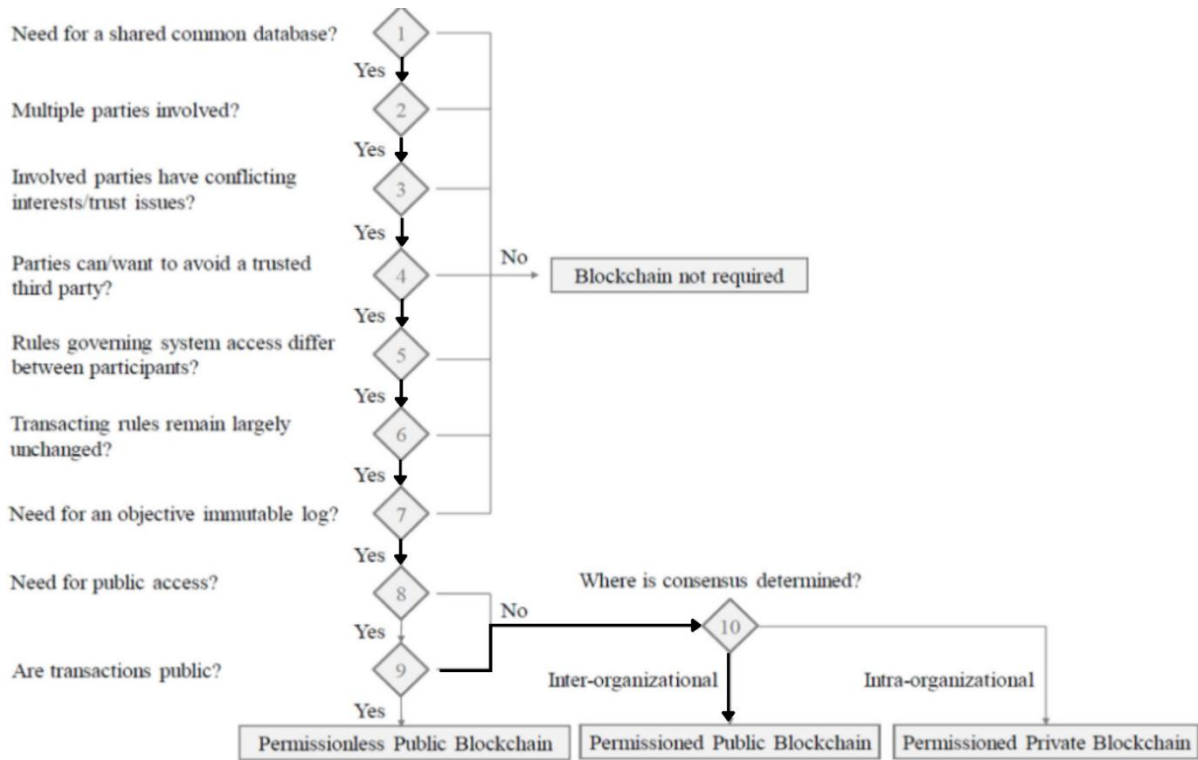


Figure 2: Ten-step blockchain suitability

For GreenHarvest’s CSA program, blockchain utilization is justified by the presence of multi-party trust issues since different independent actors have different business goals, payment delays, and product traceability demands (Lucena et al. 2018). Although internal ERP systems, permissioned databases, or RPA can automate rule-based processes and privately store data, they lack the necessary tamper-proof certification (Isbaih et al. 2024).

Additionally, a permissioned public blockchain is preferable to a public blockchain, as while transparency is important, making transaction details public could conflict with privacy laws, particularly disclosing consumers’ personal information, as mandated by the Vietnam PDPL law (Than, 2025). Therefore, some level of public accessibility for the products’ provenance, and clearly defining the roles and responsibilities of each party, are crucial for effective role-based access control to the data (Kamilaris et al. 2019).

The system architecture is further developed to illustrate the interaction and data flow within the ecosystem.

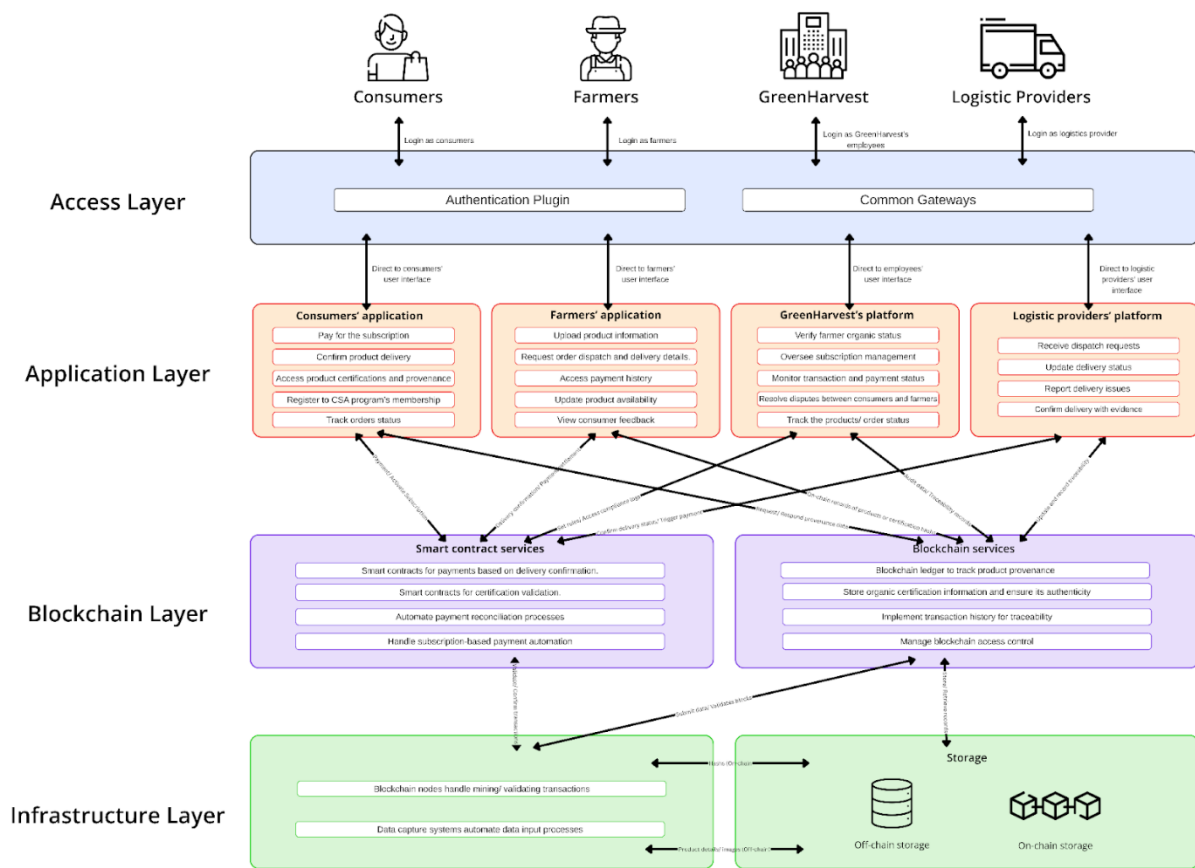


Figure 3: Data flow illustration through Blockchain-enabled System Architecture (Source: Author)
(Ellahi et al. 2023)

The system architecture consists of four layers: Access, Application, Blockchain, and Infrastructure. The Access Layer ensures role-based access control and directs the users to their corresponding roles through the authentication plugin and common gateways.

Regarding Application Layer/ Business Layer, the layer ensures the execution of business logics, where customers can order and confirm delivery, farmers upload produce information and secure payments, logistic providers can update the shipment status, and finally, GreenHarvest's cooperative monitors and controls the overall processes (Aggarwal et al. 2024; Ellahi et al. 2023).

Subsequently, the Blockchain Layer, with the smart contracts, automates reconciliation and membership subscription activation processes, and the blockchain in general will record the product details and transactions (Ellahi et al. 2023). This layer enhances the transparency and security of food traceability

Finally, the infrastructure layer will facilitate blockchain utilization through data capture tools (such as IoT sensors, gateways, or RFID), nodes, and data storage, enabling communication with blockchain ledgers (Ellahi et al. 2023). In addition to the data storage structure, the hashes will be stored on an on-

chain database, while the products' details and related images will be stored off-chain due to the data type.

III. BUSINESS PROCESS MODELING NOTATION

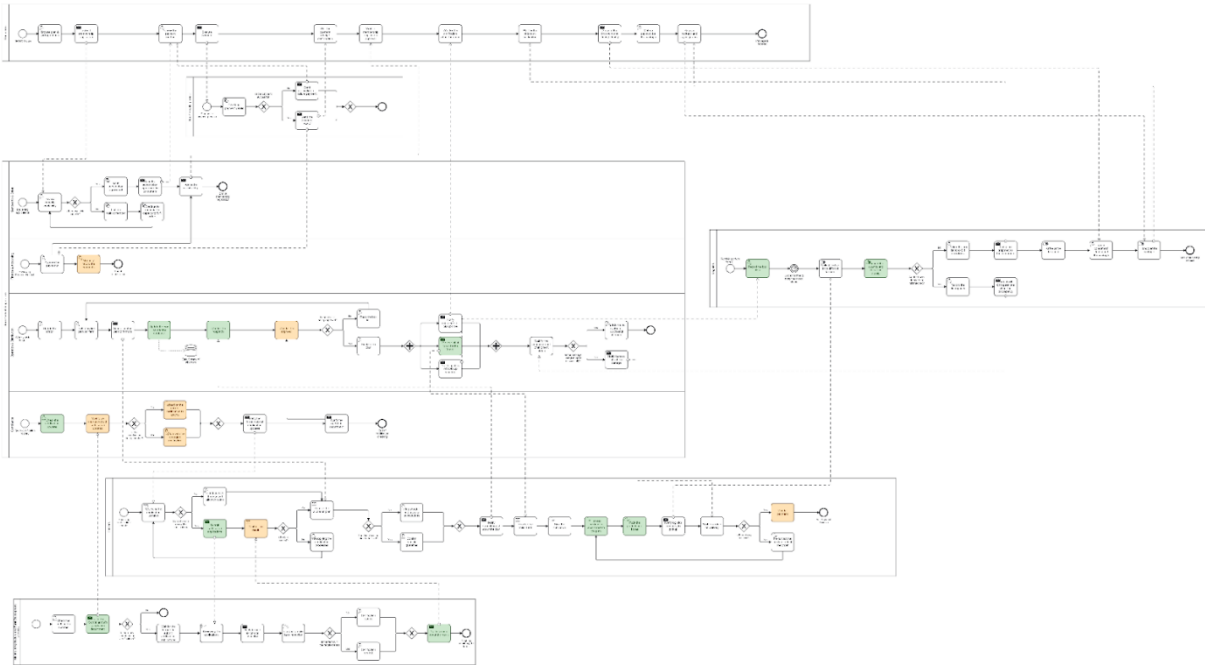


Figure 4: The GreenHarvest's CSA program order processing procedure is shown through the BPMN Diagram (Link to external source for better visualization: [The BPMN of the CSA program's order processing procedure](#))

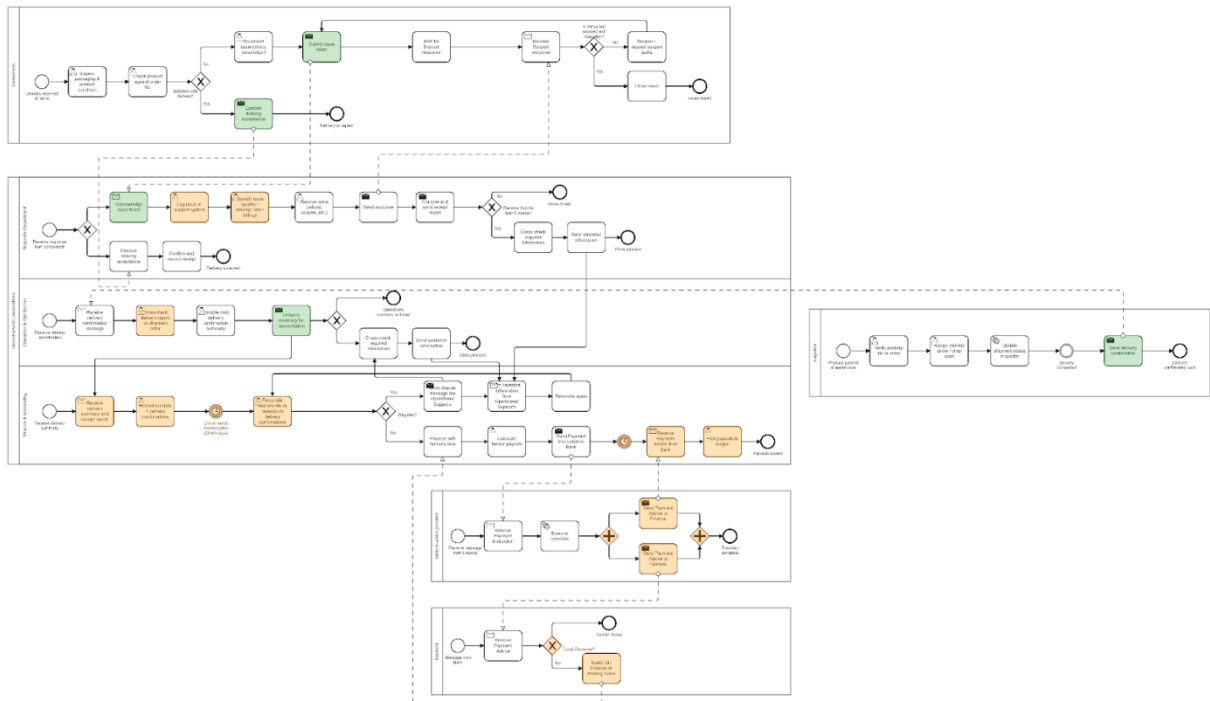


Figure 5: The GreenHarvest's CSA program payment reconciliation procedure is shown through the BPMN Diagram (Link to external source for better visualization: [The BPMN of the CSA program's payment reconciliation procedure](#))

The two BPMN diagrams provided illustrate the CSA program's workflows, covering from membership subscription, initially fund the farmers, certify processes, prepare orders (Figure 4) to finance reconciliation and farmers payment (Figure 5), with the involvement of 8 stakeholders: Consumers, Membership & Sales, Finance & Accounting, Operation & Distribution, Compliance, Certification Body, Logistics, and Bank.

Some key inefficiencies, including delayed payment due to manual reconciliation, limited product provenance traceability, and high costs caused by intermediaries, arise in tasks highlighted in green, such as cultivation without recording data, manually providing the initial funding for farmers, or waiting for certification results. These tasks will be further adjusted to fit with the blockchain adoption scope. Moreover, the tasks highlighted in yellow will be eliminated as blockchain streamlines the process.

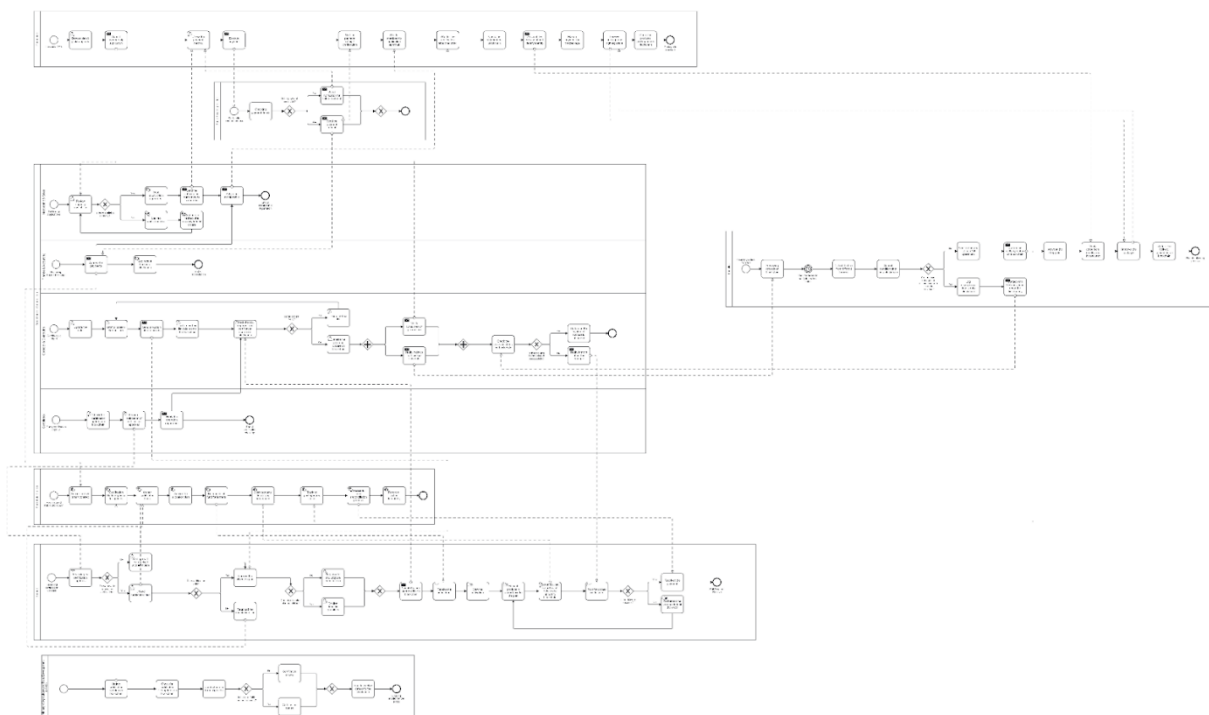


Figure 6: The GreenHarvest's CSA program order processing procedure with Blockchain being integrated is shown through the BPMN Diagram (Link to external source for better visualization: [The BPMN of the CSA program's order processing procedure with Blockchain](#))

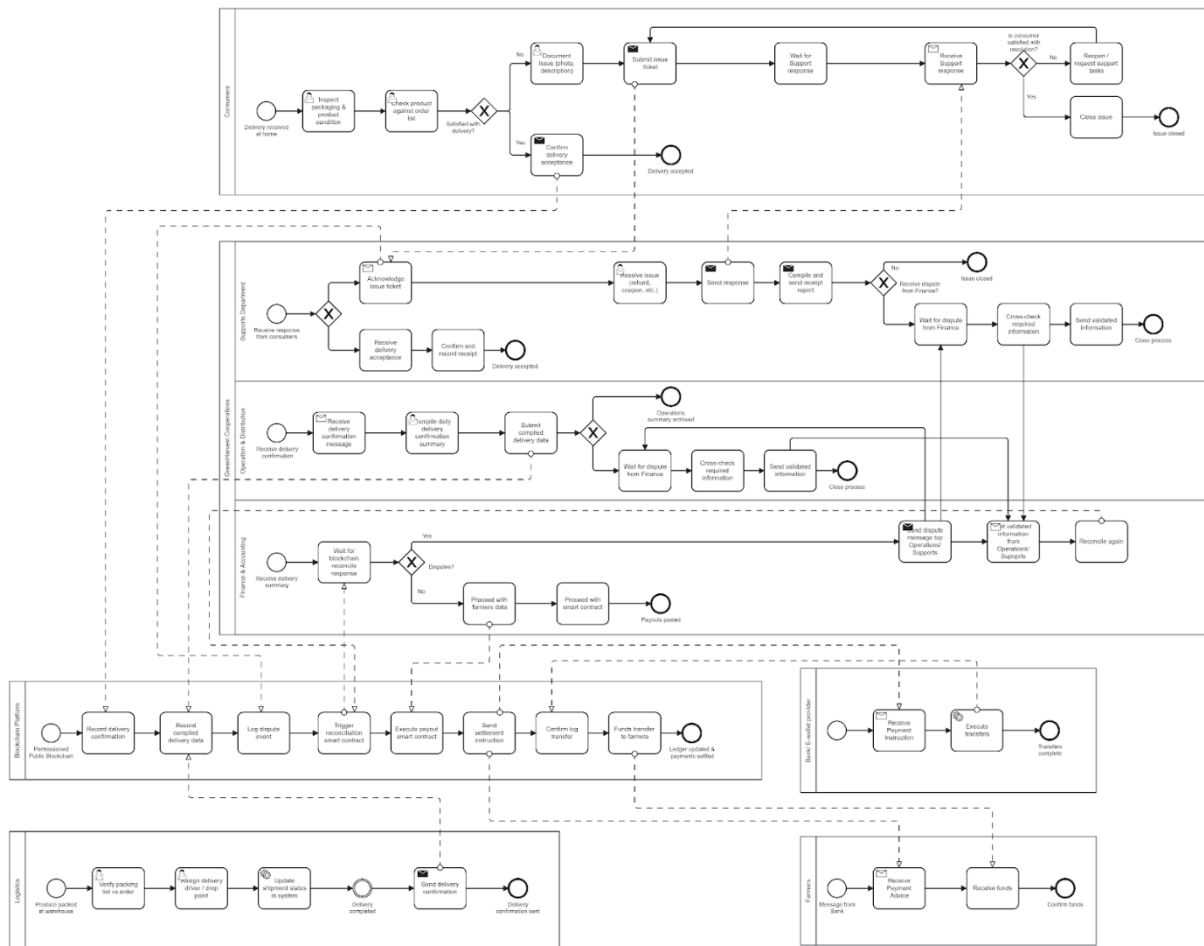


Figure 7: The GreenHarvest's CSA program payment reconciliation procedure with Blockchain being integrated is shown through the BPMN Diagram (Link to external source for better visualization: [The BPMN of CSA program's payment reconciliation procedure with Blockchain](#))

Process	Traditional BPMN	Blockchain-integrated BPMN
Payment Reconciliation	Manual reconciliation leads to delayed payments for farmers	Automated payment process after meeting certain requirements via smart contracts

Certification Verification and Product Provenance Traceability	Manual inspection process and limited visibility in the product origin cause skepticism	Real-time and immutable verification update on blockchain, from farm to table traceability
Associated Administrative Costs	Time-consuming and involves many intermediaries	Reduce time spent on tasks by eliminating redundant intermediary processes.

Table 8: Comparison table between pre- and post-integration of blockchain into GreenHarvest's operations

With the decentralized and distributed nature of blockchain, this enables all involved actors in the CSA program's process to access a shared ledger with cryptographic truth so that trade partners can interact with greater trust (Lucena et al. 2018).

Regarding efficiency, GreenHarvest Cooperative can benefit from real-time tracking and traceability, thus enabling proactive actions. For example, in Walmart's mango project, the IBM Food Trust's Hyper Ledger solution allows information regarding farm origination, batch numbers, processing data, etc., to be uploaded on an e-certificate and linked to the product package QR; Walmart significantly enhances its traceability capability, reducing the traceability time from seven days to 2.2 seconds (Casal et al. 2023; Kamath 2018).

In the case of GreenHarvest's CSA program, besides offering customers product origin data accessibility, the blockchain-based solutions are expected to execute instant payments via smart contracts instead of the 20-40 days of manual reconciliation, and also reduce 30% of overall transaction costs (Bhatia et al. 2023).

IV. STAKEHOLDER ANALYSIS AND BENEFIT

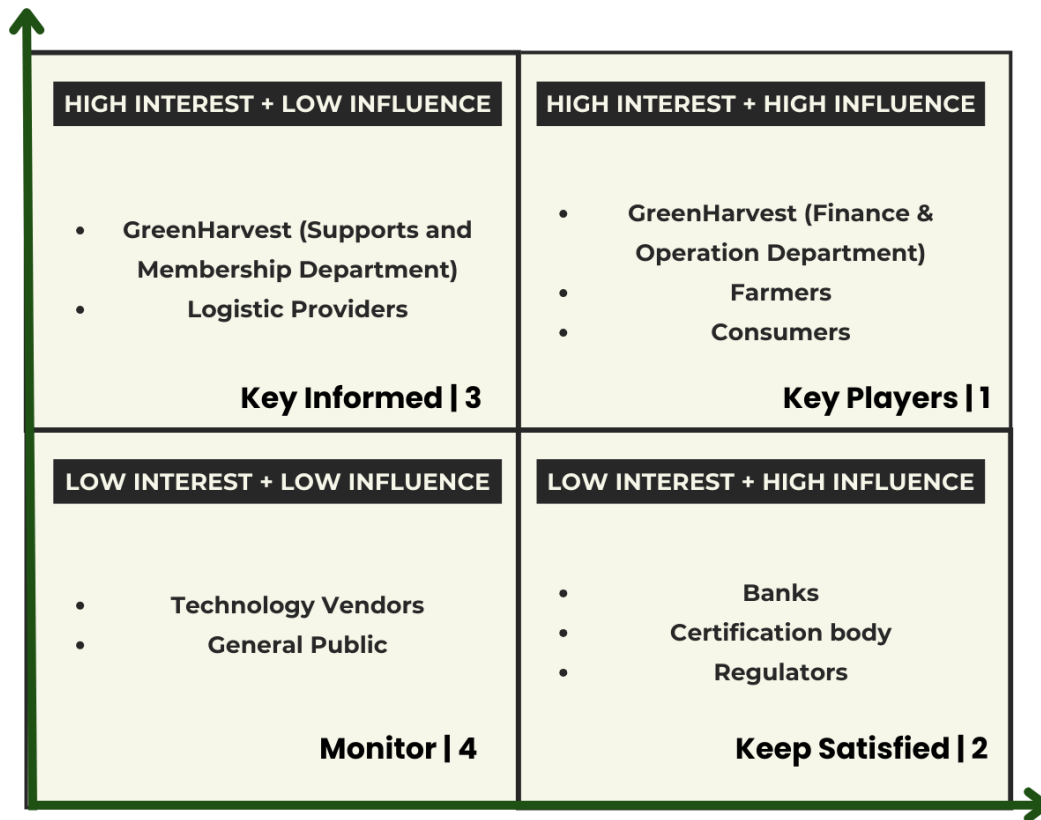


Figure 9: Stakeholder Analysis (Source: Author)

A stakeholder analysis is shown in Figure 1, with prioritization based on GreenHarvest's main challenges and the improvements blockchain can deliver. Detailed rationale tables for the two key groups, Key Players and Keep Satisfied, are included in Appendix A. Farmers, finance, and operations are Key Players as they are central to payment delays and traceability gaps. For farmers, smart contracts that trigger payment after delivery and certification can reduce the current 20-40 day delay and lower disputes (Puthenveetil and Sappati 2024).

The finance team also faces heavy reconciliation workloads and rising costs (Kamath 2018). Evidence from Walmart's blockchain pilot shows traceability reduced from days to seconds, making reconciliation time and cost ratio reliable indicators (Kamath 2018). For operations and consumers, traceability coverage and consumer trust are most relevant, since immutable provenance enhances food safety and loyalty (Kamath 2018; Tegeltija et al. 2022).

Meanwhile, the Keep Satisfied group, including banks, certification bodies, and regulators, has a strategic role. First, banks prioritize compliance and fraud prevention, so discrepancy rate and settlement time are key (Mavilia and Pisani 2019; Reuters 2024). Second, certification bodies benefit from on-chain verification, with certification cycle time and digital coverage as useful KPIs (Tegeltija et al. 2022). Finally, regulators rely on auditability and PDPD compliance, making compliance rate and inspection preparation time appropriate (TE-FOOD 2018; World Bank 2022).

V. GOVERNANCE AND CONSENSUS

For GreenHarvest's permissioned public blockchain, a Practical Byzantine Fault Tolerance (PBFT) consensus mechanism is most appropriate, as it offers fast processing, tolerates faulty nodes, and minimizes excessive energy usage. Among the platforms, Hyperledger Fabric would be used as the GreenHarvest blockchain development platform for its enterprise-grade design and proven use in a food traceability platform like IBM Food Trust (IBM 2021; Kamath 2018)

The blockchain governance model (Appendix B) is used to systematically evaluate the governance of blockchain through a combination of six dimensions and three layers (Pelt et al. 2021).

Off-chain Community	Off-chain development	On-chain protocol
ROLES		
• Farmers • Consumers • GreenHarvest's internal departments	• Advisors • Development team	• Validator nodes • Lightweight nodes: farmer's nodes, consumer's nodes
INCENTIVES		
• Farmers gain access to a reliable market, produce journey visibility, and faster payment. • Consumers are motivated by trust in the product's provenance. • GreenHarvest's departments benefit from streamlined operations and fewer manual practices.	Developers and advisors are salaried and funded by GreenHarvest's programs	All nodes benefit from access to the reliable and tamper-proof ledger that reduces fraud and errors.
MEMBERSHIP		
Open/permission community: eligible and authorized stakeholders can participate • Farmers: register and meet certain quality standards • Consumers: subscribe to membership program • GreenHarvest's departments: by employing role-based access control, certain information will be open for authorized departments based on their roles and responsibilities	Development processes are largely closed. Although the core platform using open-source, the write access and decision power belong to authorized GreenHarvest's development team. New developers can be hired or contracted by GreenHarvest	The blockchain is permissioned. • The new validator nodes can participate the blockchain network through multi-party agreement • The new lightweight nodes will receive access credentials to submit transactions or view data
COMMUNICATION		
Both through informal and formal channels • GreenHarvest can communicate with the farmers through farm visits, local meetup) or portals for sending notifications to farmers • Consumers and GreenHarvest can communicate through social media, forum, or dedicated portals for consumer	Developers coordinate through project management tools or repositories, where issues and feature proposals are tracked.	• Validator nodes communicate using a universal data diffusion model. Data is shared between every node in the network. • Lightweight nodes need a connection via a web to retrieve information about the blockchain.
DECISION MAKING		
Feedback mechanism: consumers and farmers can input their feedbacks according to their portals for further development of platform	The two formal mechanism through which developers make decisions are follow change management procedure (similar to EIP Ethereum Improvement Proposal procedure but internal) or core developer calls	The consensus mechanism of the permissioned public blockchain is Practical Byzantine Fault Tolerance (PBFT) where every new block must be agreed upon $\frac{2}{3}$ validator nodes. If protocol parameters need adjustment, a proposal need to be conducted off chain and execute on-chain.

Figure 10: GreenHarvest's blockchain governance model (Source: Author) (Pelt et al. 2021)

VI. CONCLUSION

In conclusion, GreenHarvest's CSA program can benefit from a permissioned public blockchain solution, where manual processes like reconciliation are automated, speeding up the payment process for farmers while lowering the administrative costs. However, deploying the project requires proper governance processes.

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APPENDICES

Appendix A

	Pain Points	Blockchain Benefits	KPIs (targeted measurements)
Farmers	- Delayed payments (due to manual reconciliation) - Lack of transparency (in yield/quality disputes)	- Smart contracts release funds when conditions met (delivery, certification) (<i>Puthenveetil and Sappati 2024</i>) → faster payments; immutable lot data reduces disputes	- Payment lead time: 20–40 days → few minutes to under 24h (<i>Puthenveetil and Sappati 2024</i>) - Dispute rate (whether rate decrease after estimated crop cycles)
GreenHarvest (Finance Department)	- Heavy reconciliation workload - Data mismatches across systems - High admin costs	- Event logging + instant traceability → quicker audits - Food pilots cut traceability time from days to seconds (<i>Kamath 2018</i>)	- Reconciliation time (days to minutes or seconds) (<i>Kamath 2018</i>) - Admin cost (% revenue) (<i>Kamath 2018</i>)
GreenHarvest (Operation Department)	- Lack of end-to-end visibility - Difficult logistics-certification-warehouse synchronization - Slow incident response	- On-chain farm-to-fork traceability - IoT/QR for real-time updates (<i>Demestichas et al. 2020</i>)	- Traceability coverage (<i>Kamath 2018</i>). - Time to isolate lots: seconds to minutes vs days (<i>Kamath 2018</i>).
Consumers	- Skepticism toward organic certification - Lack of provenance proof - Food safety concerns	- Blockchain-secured certification & supply chain events - Farm-to-fork pilots build trust (<i>Hilten et al. 2020; Tegeltija et al. 2022</i>)	% products with public certification: 100% of organic line (<i>Tegeltija et al. 2022</i>) - Trust proxy (trace-page views, etc) (<i>Hilten et al. 2020</i>)

Figure A1: Rationale stakeholder table for Key Players group

	Pain Points	Blockchain Benefits	KPIs (targeted measurements)
Bank	• Fraud risk • Reconciliation errors • Manual compliance (KYC/AML) • Slow fund settlement	• Smart contracts + immutable data improve payment transparency, reduce fraud, and lower audit overhead • AgriDigital shows smart-contract settlement in grains; trade finance pilots <i>(Mavilia and Pisani 2019; Reuters 2024)</i>	• Discrepancy rate • Settlement cycle time • Compliance audit readiness <i>(Mavilia and Pisani 2019; Reuters 2024)</i>
Certification Body	• Overloaded with organic verification • Slow, paper-heavy processes • fraud in organic claims	• On-chain certification + sensor data cut manual work and strengthen provenance • Studies on SAFE model and blockchain traceability confirm efficiency and fraud reduction <i>(Tegeltija et al. 2022)</i>	• Certification cycle time • % digital certificates on-chain <i>(Tegeltija et al. 2022)</i>
Regulators	• Weak oversight of food safety and labeling fraud • Limited transparent data • PDPD compliance risks.	• Permissioned blockchain enables selective access and transparent supply chain data • TE-FOOD (VN) and global pilots show large-scale support for traceability and inspection efficiency <i>(TE-Food 2018; World Bank 2022)</i>	• Regulatory compliance rate • Inspection preparation time • PDPD-compliant <i>(TE-Food 2018; World Bank 2022)</i>

Figure A2: Rationale stakeholder table for Keep Informed group

Appendix B



Figure B1: Blockchain governance framework definition (Source: Pelt et al. 2021)